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(54) **SYSTEM AND METHOD FOR TOKENIZING OFFERS**

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(58) **Field of Classification Search**

None

See application file for complete search history.

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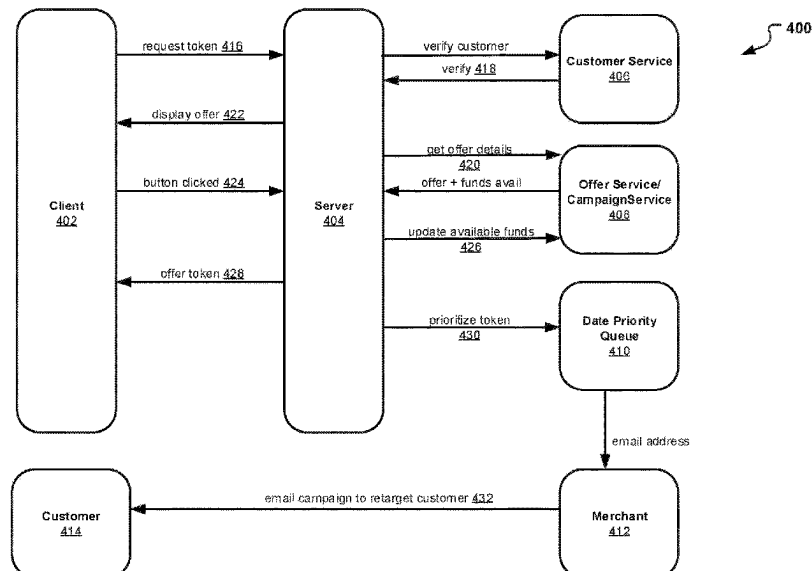
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(57) **ABSTRACT**

Aspects of the present disclosure involve systems, methods, devices, and the like for tokenizing offers. The current disclosure presents a system and method that can present a tokenized offer for saving and retrieving from a digital wallet. The tokenized offer may be presented on a client site for selection by a user interested in taking advantage of the promotion without having to leave the current site. The current disclosure also presents a system that can retrieve the tokenized offer saved for use with a purchase. The tokenized offer may be automatically applied and on display at the merchant site and/or retrieved from the digital wallet.

20 Claims, 7 Drawing Sheets



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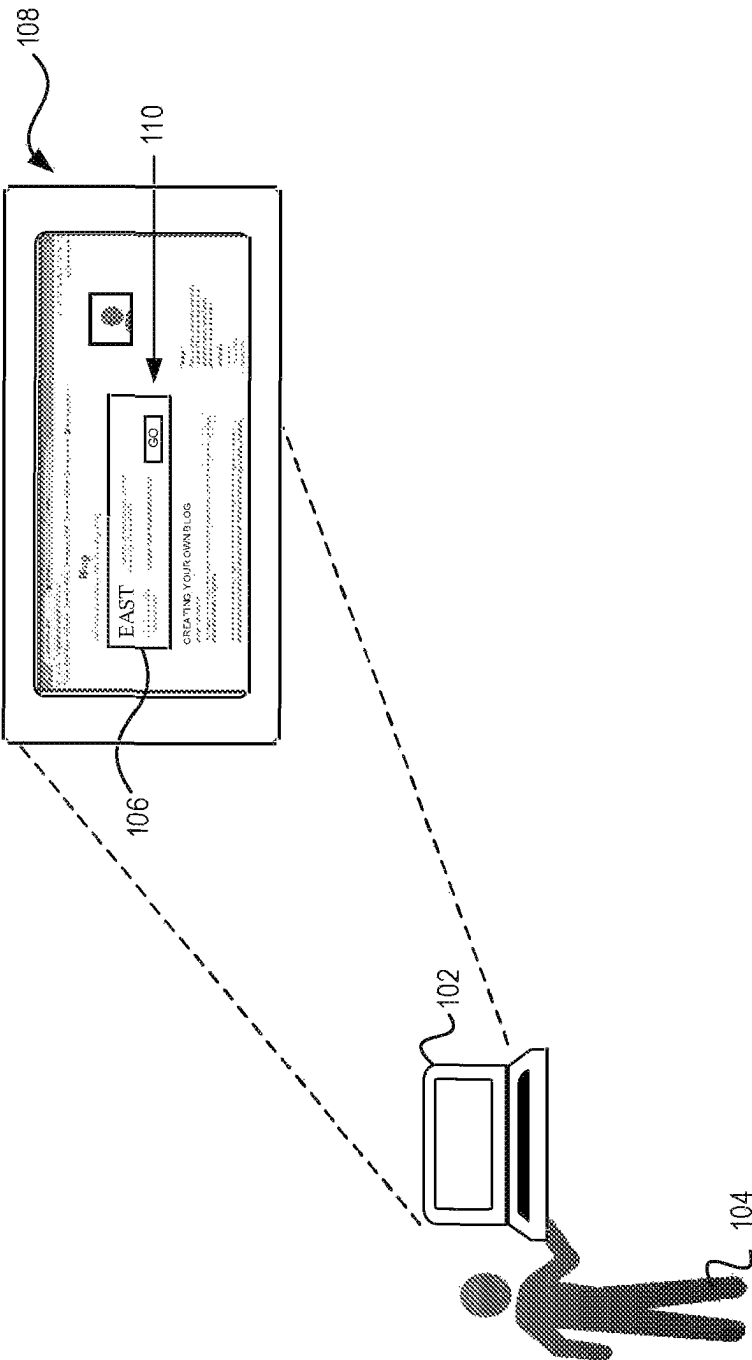


FIG. 1

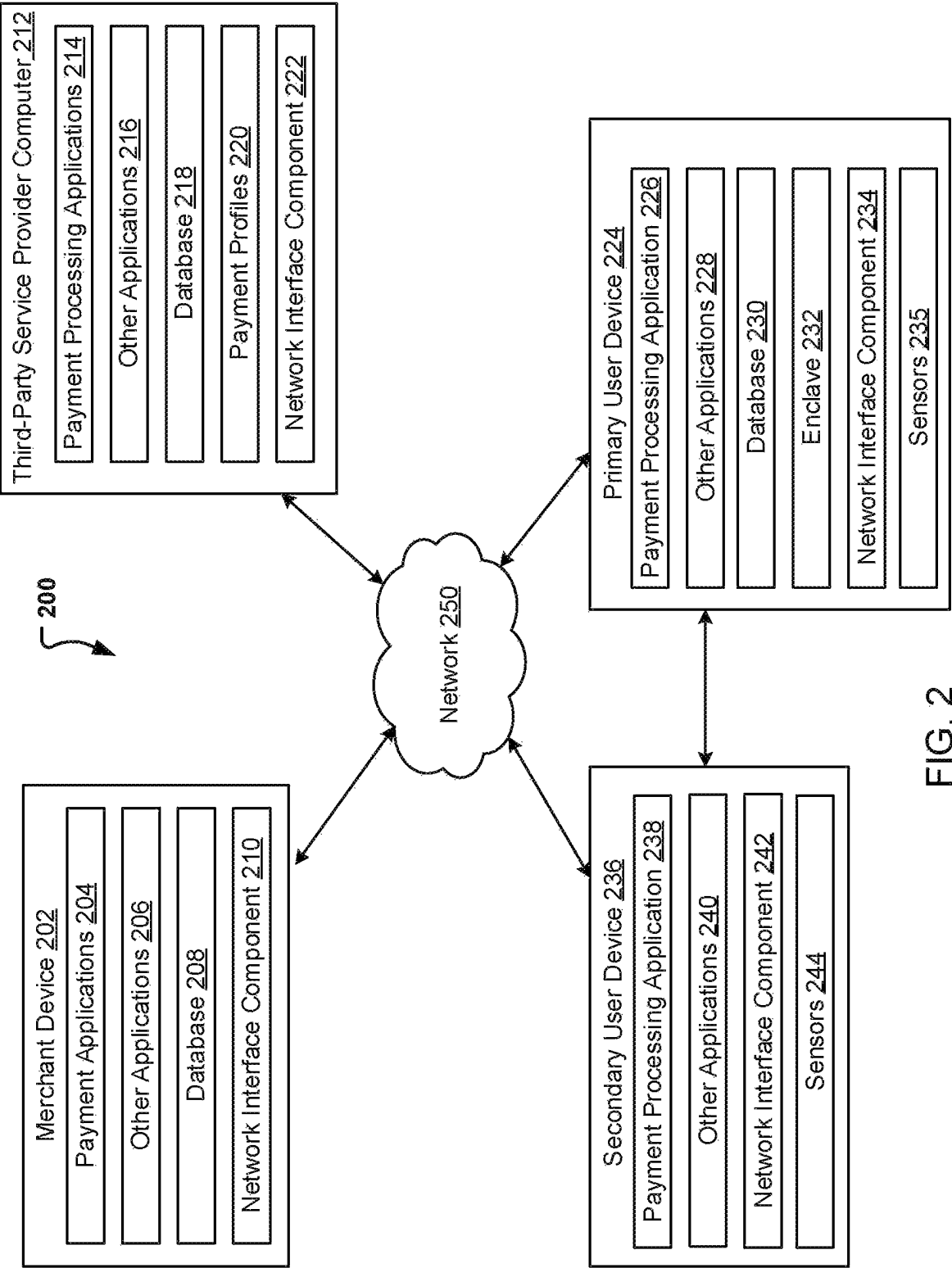


FIG. 2

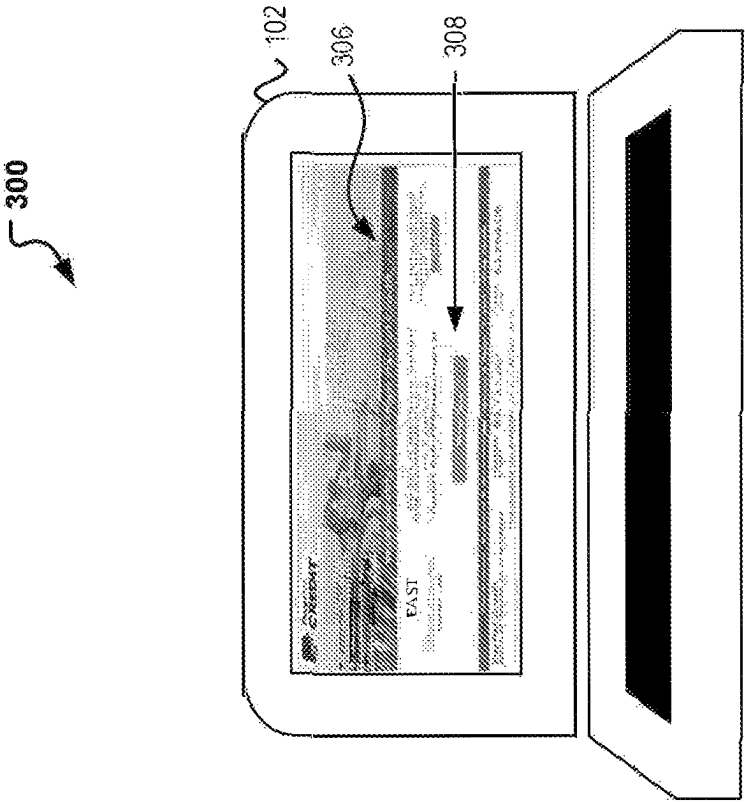


FIG. 3A

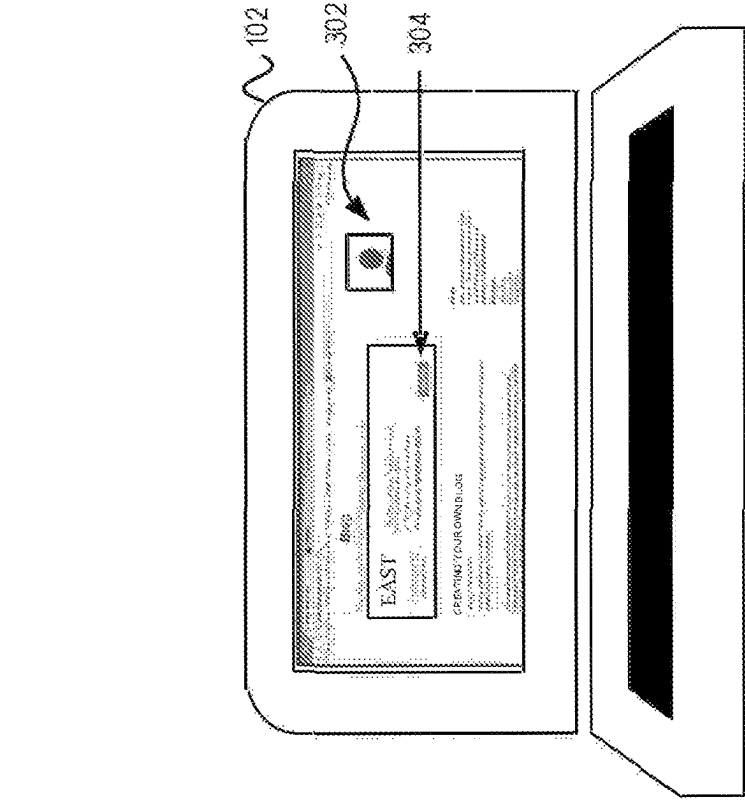


FIG. 3B

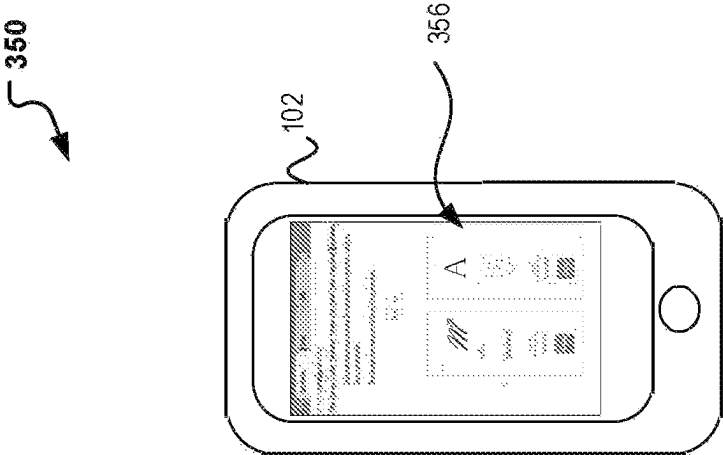


FIG. 3D

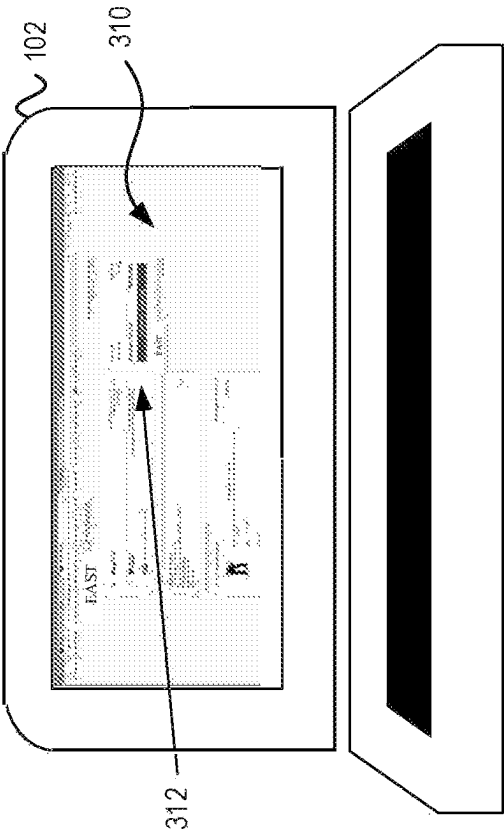


FIG. 3C

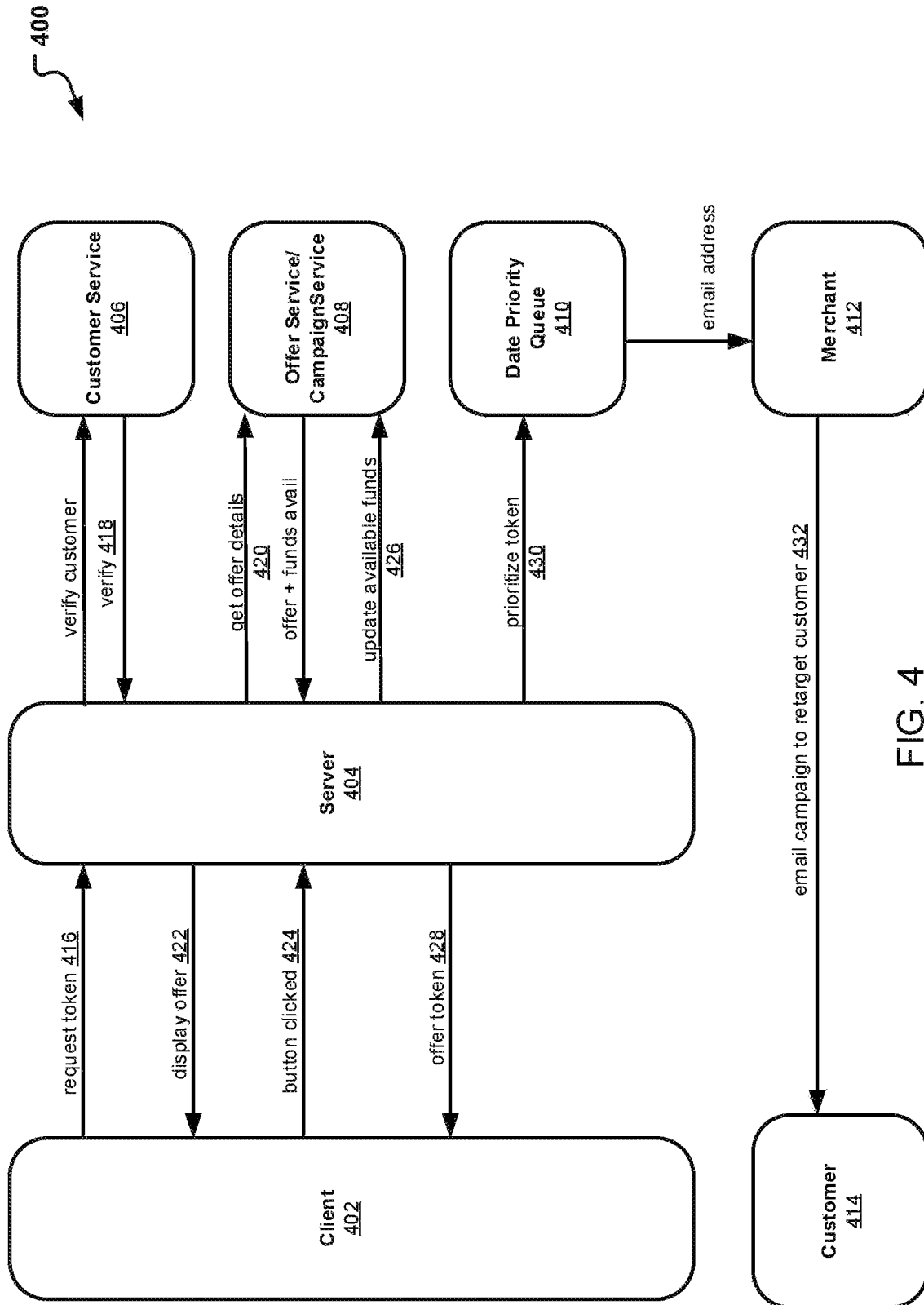


FIG. 4

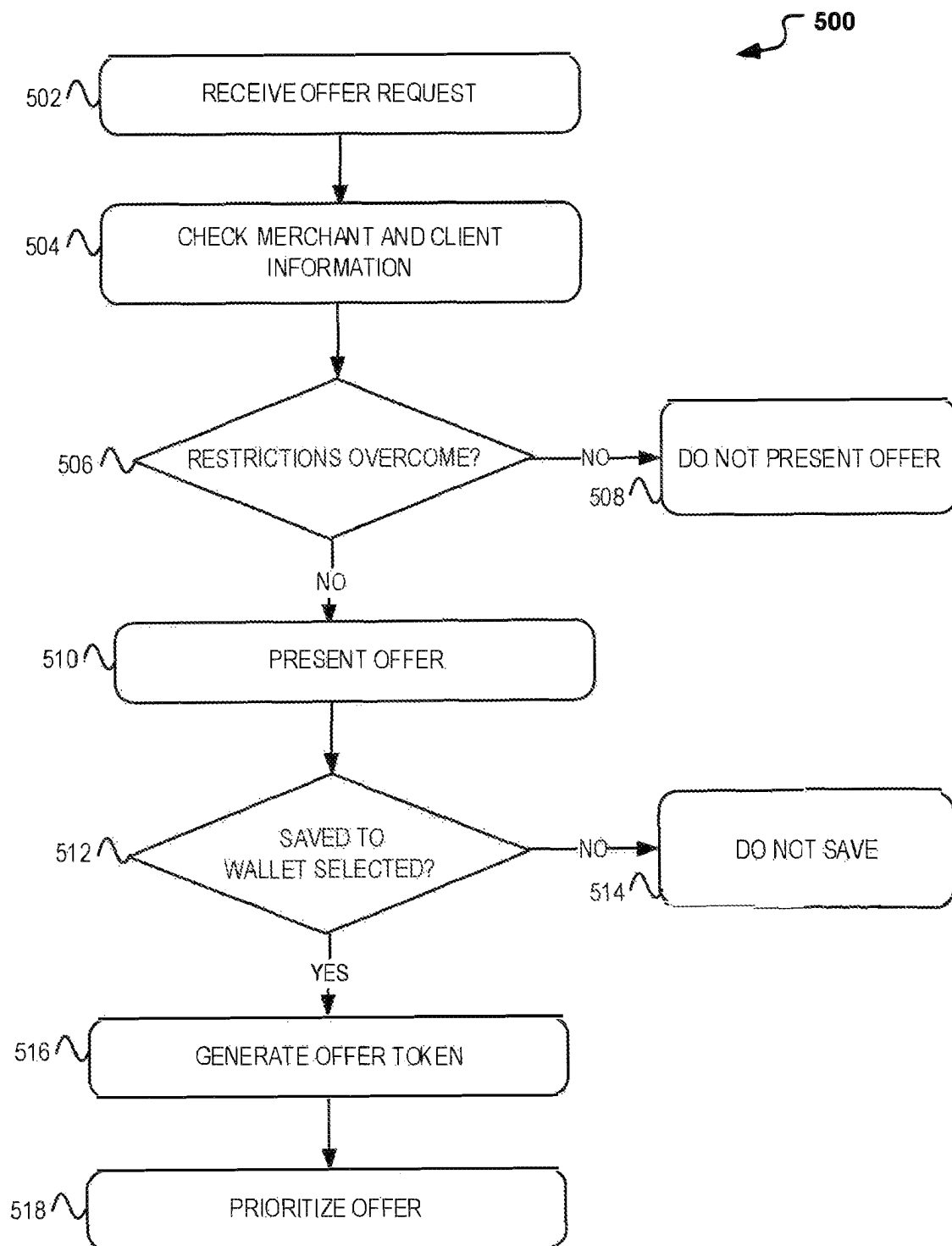


FIG. 5

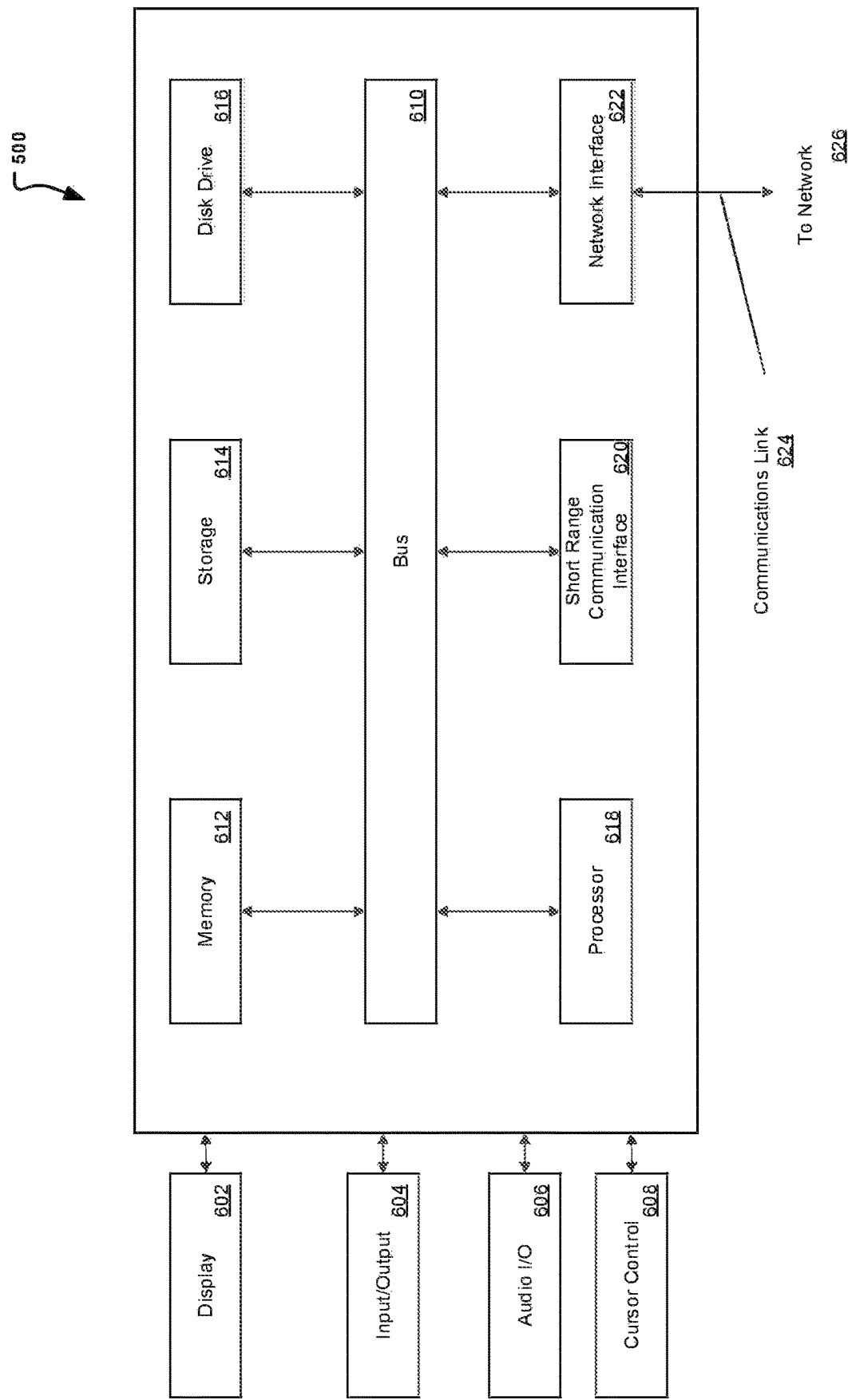


FIG. 6

SYSTEM AND METHOD FOR TOKENIZING OFFERS

RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/336,209, filed Jun. 1, 2021, which is a continuation of U.S. application Ser. No. 15/695,473, filed Sep. 5, 2017, now U.S. Pat. No. 11,023,929, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

The present disclosure generally relates to communications between devices, and more specifically, to the communication between devices for receiving tokenized offers.

BACKGROUND

Rapid advancements in technology and communications have led to the ubiquitous use of wireless electronics. As such, today's society has grown to become heavily device reliant, with a need to be constantly connected. This reliance has led to the use of electronics to make monetary transactions for goods and services.

For example, consumers can now go to a home computer to purchase an item at an online store site. In some instances, the user can obtain special offers and promotions from these online stores while browsing a different site. Unfortunately, conventional systems generally require the user to leave the current site in order to obtain and use the promotion. However, a user may have an interest in remaining on the current site to continue reading, browsing, streaming, or other activity, at the moment. Thus, the promotion presented to the user, may be lost and possible sale for the merchant postponed or not completed. This loss of sale, offer, or promotion can lead to a loss of time (if redirected), money, as well as stress and frustration to a user. Therefore, it would be beneficial if a system were created that provides user with the ability to store and access such promotions at a later time.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 illustrates a block diagram of a first user receiving an offer while interacting with one or more devices.

FIG. 2 illustrates a block diagram of a system for tokenizing offers.

FIGS. 3A-3D illustrate a communication with a user device for obtaining tokenized offers.

FIG. 4 illustrates a block diagram of communication between devices for tokenizing offers.

FIG. 5 illustrates a flow diagram for tokenizing offers.

FIG. 6 illustrates an example block diagram of a computer system suitable for implementing one or more devices of the communication systems of FIGS. 2 and 5.

Embodiments of the present disclosure and their advantages are best understood by referring to the detailed description that follows. It should be appreciated that like reference numerals are used to identify like elements illustrated in one or more of the figures, whereas showings therein are for purposes of illustrating embodiments of the present disclosure and not for purposes of limiting the same.

DETAILED DESCRIPTION

In the following description, specific details are set forth describing some embodiments consistent with the present

disclosure. It will be apparent, however, to one skilled in the art that some embodiments may be practiced without some or all of these specific details. The specific embodiments disclosed herein are meant to be illustrative but not limiting.

One skilled in the art may realize other elements that, although not specifically described here, are within the scope and the spirit of this disclosure. In addition, to avoid unnecessary repetition, one or more features shown and described in association with one embodiment may be incorporated into other embodiments unless specifically described otherwise or if the one or more features would make an embodiment non-functional.

Aspects of the present disclosure involve systems, methods, devices, and the like for tokenizing offers. In one embodiment, a system is introduced that can present a tokenized offer for saving and retrieving from a digital wallet. The tokenized offer may be presented on a client site for selection by a user interested in taking advantage in the promotion without having to leave the current site. The tokenized offer may be presented in response to a verification that a customer meets restrictions and/or criteria set by a merchant. In another embodiment, a system is introduced that can retrieve the tokenized offer saved for use with a purchase. The tokenized offer may be automatically applied and on display at the merchant site and/or retrieved from the digital wallet.

Rapid advances in communications have made portable electronic devices, such as smartphones and tablets, part of everyday life. Among other things, these electronic devices can be used to browse the web and purchase goods and services. In some instances, the user may use the electronic device to browse the web or stream video, while in other instances, the user may be specifically looking to purchase an item. Oftentimes, however, a promotion for an item may appear at an inopportune time, when the user is not specifically searching and instead reading an article. In these instances, the user may find it difficult to leave the current site in order to take advantage of the current promotion presented.

An example of this is illustrated in FIG. 1, where a user 104 is interacting with a user device 102. In particular, FIG. 1 illustrates a block diagram 100 of a user 104 receiving an offer 106 while interacting with one or more devices 102. The user device 102 can be a smart phone, tablet, laptop, and other electronic device that may be used for communicating, browsing, purchasing, and the like.

For illustration purposes, a user 104 is presented interacting with a user device 102, engaged in learning how to create a blog from a site 108. The site 108 may include user profile information (e.g., user name and picture), instructions, and other details relevant to the current site 108 visited. In addition, the site 108, may also include advertisements, news articles, and even offers and promotions in the form of a coupon 106. The coupon may be available for use at a different site, with another merchant, after a purchase, etc. Oftentimes, to take advantage of the coupon, the user will be redirected to another site corresponding to the entity offering the discount. As illustrated in FIG. 1, the coupon 106 may be accompanied with a button 110 for redirecting the user 104. However, as indicated, the user may be in the middle of learning a new skill set (e.g., learning how to create a blog) and may not be not interested in leaving the current site. Thus, the user 104 may be left with a decision to forgo the saving presented in order to continue on the current site 108. Therefore, in order to eliminate the need to relinquish such promotion, tokenized offers are

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introduced that can provide a system and method for saving and accessing such coupon **110** at a later time.

In the current embodiment, a system is presented that is able to communicate with a client member offering the promotion to generate a tokenized offer for later use. The tokenized offer is stored using a token that may be generated in response to an offer acceptance by the user and available for accessing, viewing, and using at a later time. In one embodiment, the tokenized offer may be saved to a digital wallet and/or with a payment provider for later use.

FIG. **2** is a block diagram of a networked system **200** for implementing the processes described herein, according to an embodiment. In particular, FIG. **2** illustrates a block diagram of a system **200** for tokenizing offers. System **200** illustrates at least some of the interactions between multiple entities and/or devices for the authentication, use and purchase of a product or service with a merchant in association with a service provider. As shown, system **200** may include or implement a plurality of devices, computers, servers, and/or software components that operate to perform various methodologies in accordance with the described embodiments. It will be appreciated that the devices, computers, and/or servers illustrated in FIG. **2** may be deployed differently and that the operations performed and/or the services provided by such devices, computers, and/or servers may be combined or separated for a given embodiment and may be performed by a greater number or fewer number of devices, computers, and/or servers. Furthermore, one or more of the devices, computers, and/or servers may be operated and/or maintained by the same or different entities.

System **200** includes a merchant device **202**, a primary user device **224** (e.g., primary device/trusted primary device **102**), a third-party service provider computer **212**, and a secondary user device **236** (e.g., secondary device **104**) in communication over a network **250**. The merchant device **202**, primary user device **224**, third-party service provider computer **212**, and the secondary user device **236** may each include one or more processors, memories, and other appropriate components for executing computer-executable instructions such as program code and/or data. The computer-executable instructions may be stored on one or more computer readable mediums or computer readable devices to implement the various applications, data, and steps described herein. For example, such instructions may be stored in one or more computer readable media such as memories or data storage devices internal and/or external to various components of system **200**, and/or accessible over network **250**.

The merchant device **202** may be implemented as a communication device that may utilize appropriate hardware and software configured for wired and/or wireless communication with the primary user device **224**, third-party service provider computer **212**, and/or secondary user device **236**. For example, the merchant device **202** may be implemented as a personal computer (PC), a smart phone, laptop/tablet computer, point-of-sale device, wristwatch with appropriate computer hardware resources, eyeglasses with appropriate computer hardware, other type of wearable computing device, implantable communication devices, and/or other types of computing devices capable of transmitting and/or receiving data. The merchant device **202** may correspond to and be utilized by a user, such as an employee of a merchant and/or another person authorized by the merchant.

The merchant device **202** may include one or more payment applications **204**, other applications **206**, a database **208**, and a network interface component **210**. The payment

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applications **204** and other applications **206** may correspond to executable processes, procedures, and/or applications with associated hardware. In other embodiments, merchant device **202** may include additional or different components having specialized hardware and/or software to perform operations associated with the payment applications **204** and/or the other applications **206**.

The payment application **204** may facilitate financial transactions corresponding to the sale of goods and/or services offered by the merchant. For example, the payment application **204** may provide an interface for customers to purchase the goods or services and to receive customer payment information (e.g., customer credit card information). The payment application **204** may further transmit customer payment information to a payment processor (e.g., such as a payment processor corresponding to the third-party service provider computer **212** or the secondary user device **236**) to process the customer payment information. The payment application **204** may also facilitate other types of financial transactions such as banking, online payments, money transfer, discount coupon information, and/or the like.

The merchant device **202** may execute the other applications **206** to perform various other tasks and/or operations corresponding to the merchant device **202**. For example, the other applications **206** may include security applications for implementing client-side security features, programmatic client applications for interfacing with appropriate application programming interfaces (APIs) over network **250**, or other types of applications. The other applications **206** may also include additional communication applications, such as email, texting, voice, and instant messaging (IM) applications that enable a user **106** to send and receive emails, calls, texts, and other notifications through the network **250**. In various embodiments, the other applications **206** may include location detection applications, such as a mapping, compass, and/or global positioning system (GPS) applications, which may be used to determine a location of the merchant device **202**. The other applications may **206** include social networking applications. Additionally, the other applications **206** may include device interfaces and other display modules that may receive input and/or output information. For example, the other applications **206** may include a graphical user interface (GUI) configured to provide an interface to the user.

The merchant device **202** may further include a database **208**, which may be stored in a memory and/or other storage device of the merchant device **202**. The database **208** may include, for example, identifiers (IDs) such as operating system registry entries, cookies associated with the payment application **204**, tokens, encryption keys, and/or other applications **206**, IDs associated with hardware of the network interface component **210**, IDs used for payment/user/device authentication or identification, and/or other appropriate IDs. The database **208** may also include information corresponding to one or purchase transactions of customers who have purchased goods or services from the merchant, browsing histories of the customers, or other types of customer information. In certain embodiments, the merchant device **202** may also include information corresponding to payment tokens, such as payment tokens generated by the third-party service provider computer **212** and/or generated by the secondary user device **236**.

The merchant device **202** may also include at least one network interface component **210** configured to communicate with various other devices such as the primary user device **224**, the third-party service provider computer **212**,

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and/or the secondary user device **236**. In various embodiments, network interface component **210** may include a Digital Subscriber Line (DSL) modem, a Public Switched Telephone Network (PSTN) modem, an Ethernet device, a broadband device, a satellite device and/or various other types of wired and/or wireless network communication devices including microwave, radio frequency, infrared, Bluetooth®, Bluetooth low-energy, near field communication (NFC) devices, and/or the like.

The third-party service provider computer **212** may be maintained, for example, by a third-party service provider, which may provide payment processing services for the merchant. In one example, the third-party service provider may be provided by PAYPAL®, Inc. of San Jose, CA, USA. Alternatively, the third-party service provider computer **212** may be associated with a user of the primary and secondary device **224**, **236**. As such, the third-party service provider computer **212** includes one or more payment processing applications **214**, which may be configured to process payment information received from the merchant device **202** or from a selection at the primary or secondary user device **224**, **236**. For example, the payment application **204** of the merchant device **202** may receive payment information from a customer to purchase a service or good offered by the merchant. Upon receipt of the payment information, the payment application **204** may transmit the payment information to the third-party service provider computer **212**. The payment processing application **214** of the third-party service provider computer **212** may receive and process the payment information. As another example, the payment application **204** can present a payment code on a display of the user device associated with the merchant. The payment code can be scanned or transmitted to the merchant device **202** for payment processing.

The third-party service provider computer **212** may execute the other applications **216** to perform various other tasks and/or operations corresponding to the third-party service provider computer **212**. For example, the other applications **216** may include security applications for implementing client-side security features, programmatic client applications for interfacing with appropriate APIs over the network **250**, or other types of applications. The other applications **216** may also include additional communication applications, such as email, texting, voice, and IM applications that enable communication of emails, calls, texts, and other notifications through the network **250**. In various embodiments, the other applications **216** may include location detection applications, such as a mapping, compass, and/or GPS applications, which may be used to determine a location of the third-party service provider computer **212**. Additionally, the other applications **216** may include device interfaces and other display modules that may receive input and/or output information. For example, the other applications **216** may include a GUI configured to provide an interface to one or more users.

The third-party service provider computer **212** may further include a database **218**, which may be stored in a memory and/or other storage device of the third-party service provider computer **212**. The database **218** may include, for example, IDs such as operating system registry entries, cookies associated with the payment processing application **214** and/or other the applications **216**, IDs associated with hardware of the network interface component **222**, IDs used for payment/user/device authentication or identification, and/or other appropriate IDs.

According to a particular embodiment, the third-party service provider computer **212** may include a set of payment

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profiles **220** corresponding to past sales transactions executed by the merchant device **202** with respect to one or more customers of the merchant. Alternatively, the third-party service provider computer **212** may include a set of merchant payment profiles corresponding to the payment sources associated to a corresponding merchant. For example, a particular payment profile from the set of payment profiles **220** may include payment information corresponding to a particular customer of the merchant and/or a merchant associated with a user. The payment information may include credit card information (e.g., card number, expiration date, security code, card issuer, and/or the like), Automated Clearing House (ACH) information (e.g., account number, routing number, and/or the like), identification information associated with the particular customer/user (e.g., a customer identifier, name, address, phone number, date of birth, and/or the like), billing information, and/or any other type of payment information associated with the particular customer. Furthermore, other payment profiles of the set of payment profiles **220** may include payment information corresponding to other customers of the merchant and/or other merchants associated with the user. In addition, the third-party service provider computer **212** may store the set of payment profiles **220** according to a first file format.

The third-party service provider computer **212** may also store a set of payment tokens corresponding to the set of payment profiles **220**. For example, each payment profile of the set of payment profiles **220** may be associated with a corresponding payment token from the set of payment tokens. In some embodiments, each payment profile may include a corresponding payment token from the set of payment tokens. The set of payment tokens may be particular to the third-party service provider computer **212** (e.g., computers from other service providers may be unable to use the set of payment tokens) and may enable the merchant device **202** to more securely process payment transactions with the third-party service provider computer **212**. For example, in order to process a payment transaction that involves a credit card number associated with a particular payment profile, the third-party service provider computer **212** may provide the merchant device **202** with a particular payment token that is different from the credit card number. The merchant device **202** may use the particular payment token to process the payment transaction instead of the credit card number. Further, the merchant device may store and associate the particular payment token with the particular payment profile instead of the credit card number, thereby protecting the credit card number from being stolen in a potential security breach of the merchant device **202**.

In various embodiments, the third-party service provider computer **212** also includes at least one network interface component **222** that is configured to communicate with the merchant device **202**, the primary user device **224**, and/or the secondary user device **236** via the network **250**. Further, the network interface component **222** may comprise a DSL modem, a PSTN modem, an Ethernet device, a broadband device, a satellite device and/or various other types of wired and/or wireless network communication devices including microwave, RF, and IR communication devices.

The primary user device **224** may be implemented as a communication device that may utilize appropriate hardware and software configured for wired and/or wireless communication with the merchant device **202**, third-party service provider computer **212**, and/or the secondary user device **236**. The primary user device **224**, may be a personal computer (PC), a smart phone, laptop/tablet computer, wrist-

watch with appropriate computer hardware resources, eyeglasses with appropriate computer hardware (e.g. GOOGLE GLASS®), other type of wearable computing device, implantable communication devices, and/or other types of computing devices capable of transmitting and/or receiving data. In one embodiment, the primary user device **224** may be mobile device **102** communicating with second device **104**, a merchant device and/or service provider **212**.

The primary user device **224** may include a payment processing application **226** that may be used as a digital wallet that can communicate with a merchant device **202**, secondary user device **236**, and/or third party service provider **212** for purchasing and transacting. The payment processing application **226**, can work jointly with database **230** for retrieving bank account information, user accounts, security codes, tokens that may be associated with various merchant locations. Similarly, the payment processing application, can also provide access to the user profiles for determining which payment method, processing code, to use at a merchant location. The primary user device **224** may include an enclave **232** which can be a secure hardware level storage are designated for the payment processing application **226**. The enclave **232** can enable the storage of private and public keys and a randomly generated token that enables the secure communication and validation of the user of the payment processing application with the third party service provider **212**.

In one embodiment, the primary device **102** uses a private key and secure token encryption for communicating with the payment processing application **226**. The secure token can be pushed in a silent tone in an out of band notification between the payment processing application (e.g., PayPal application) and a server. In other words, when the payment processing application **226** is first installed, two keys are generated, a private key and a public key. The private key is put in the secure enclave **232** and the public key is sent to the database **218** or other server associated with the third-party service provider **212**. Therefore, when a user associated with the primary user device **224**, first logs into the payment processing application **226**, if the login is successful a server associated with the third-party service provider **212** or even a merchant device **202** initiates a silent tone notification to the payment processing application **226**. The push notification can be an out-of-band notification which ensures a secure access through the use of the silent tone notification including the transmission of a token. Thus, a request for access from the payment processing application **226**, should include the token encrypted which can be decrypted with the public key that was originally transmitted.

The primary user device **224** may also include other applications **228** to perform various other tasks and/or operations corresponding to the primary user device **224**. For example, the other applications **228** may facilitate communication with the merchant device **202**, such as to receive an indication, from the merchant device **202**, to switch payment processing services from the third-party service provider to the service provider. As another example, the other applications **228** may include security applications, application that enable designation of a primary interactive device, and applications that allow for web site searches (including access to merchant websites). The other applications **228** may also include additional communication applications, such as email, texting, voice, and IM applications that enable communication of emails, calls, texts, and other notifications through the network **250**. In various embodiments, the other applications **228** may include location

detection applications, such as a mapping, compass, and/or GPS applications, which may be used to determine a location of the user device **224**. The other applications may **228** social networking applications. Additionally, the other applications **228** may include device interfaces and other display modules that may receive input and/or output information. For example, the other applications **228** may include a GUI configured to provide an interface to one or more users.

The primary user device **224** may further include a database **230**, which may be stored in a memory and/or other storage device of the primary user device **224**. The database **230** may include, for example, identifiers (IDs) such as operating system registry entries, cookies associated with a web browser and/or the other applications **228**, IDs associated with hardware of the network interface component **234**, IDs used for payment/user/device authentication or identification, bank information, merchant information, user accounts, and/or other appropriate IDs. In addition, the database **230** may include user identifier information which can be used in conjunction with secondary user device **236** during an explicit code exchange and/or contextual information for web browser authentication.

Primary user device **224**, can also be equipped with various sensors **234** to provide sensed characteristics about the primary user device **224** and its environment. For example, the sensors **234** can include a camera for taking images of the surroundings used to determine the information to provide a user. As another example, the sensors **234** can include microphones for receiving audio signals, which can be used to detect location, and other relevant terms that may be spoken. Further, the sensors **234** can include an accelerometer, a light sensor, a biometric sensor, temperature sensor, etc. For example, the biometric sensor can be used to collect a user fingerprint scan. Each type of sensor providing various characteristics about the device, user, or environment.

The primary user device **224** may also include at least one network interface component **110** configured to communicate with various other devices such as the merchant device **202**, the third-party service provider computer **212**, and/or the secondary user device **236**. In various embodiments, network interface component **234** may include a Digital Subscriber Line (DSL) modem, a Public Switched Telephone Network (PTSN) modem, an Ethernet device, a broadband device, a satellite device and/or various other types of wired and/or wireless network communication devices including microwave, radio frequency, infrared, Bluetooth®, Bluetooth low-energy, near field communication (NFC) devices, and/or the like.

As indicated above, a secondary user device **236** (e.g., secondary device **104**) may be presently available and used by a consumer (e.g., user **106**). The secondary user device **236**, much like the primary user device may be equipped with payment processing applications **238** as well as other applications **240**. The payment processing applications **238** and other applications **240** may be used to perform various tasks and/or operations corresponding to the secondary user device **236**. The other applications **228** may include additional communication applications, such as email, texting, voice, and IM applications that enable communication of emails, calls, texts, and other notifications through the network **250**. In addition, the other applications **228** may include location detection applications, such as a mapping, compass, and/or GPS applications, which may be used to determine a location of the secondary user device **236**. The other applications may further include **228** social network-

ing applications and device interfaces and other display modules that may receive input and/or output information.

For example, the applications **238**, **240** may facilitate communication with the merchant device **202**, such as by presenting a QR code that may be used for making a purchase. As another example, the applications **238**, **240** may provide notice of an incoming call or text to the user **106** of the secondary user device **236**. Still as another example, the applications **238**, **240** can provide details and/or login information for authentication by the primary user device **224**, with a merchant device **202**, or third-party service provider **212**.

The secondary user device **236** may also include a network interface component **242** for connecting and interacting with at least primary user device **224**, merchant user device **202**, and/or third-party service provider computer **212** over network **250**. The network interface component **242** may comprise a DSL modem, a PSTN modem, an Ethernet device, a broadband device, a satellite device and/or various other types of wired and/or wireless network communication devices including microwave, RF, and IR communication devices.

The network **250** may be implemented as a single network or a combination of multiple networks. For example, in various embodiments, the network **250** may include the Internet or one or more intranets, landline networks, wireless networks, and/or other appropriate types of networks. Thus, the network **250** may correspond to small scale communication networks, such as a private or local area network, or a larger scale network, such as a wide area network or the Internet, accessible by the various components of system **200**.

In addition, the secondary user device **236** can also include sensors **244** that may be used for providing measured information about the user, device, location, environment, etc. The sensors can be used to detect light (is device in a pocket), biometric data (is user working out), location (user is in a meeting, call received), etc. Further, the sensors **244** can also be used to determine proximity between devices and location of interest as well as determine processor usage and/or signal strength relative to other devices including primary user device **224**, third party service provider computer **212**, and merchant device **202**.

Further, system **200** can also include a communication between the secondary user device **236** and/or primary user device **224** with third-party service provider computer **212** for saving and retrieving offers tokenized and stored for later use. To illustrate this communication, FIGS. **3A-3D** are presented where a user device **102** used in obtaining tokenized offers. As an example, a third party service provider (e.g., PayPal) will be used store and retrieve the offers. In addition, various offers from numerous merchants will be described however, the details are not so limited as other merchants, websites, discounts, promotions, and service providers may be contemplated.

FIG. **3A** begins with an illustration of a user **102** browsing a site **302** that describes how to create a blog. The site **302** can include, among other things, instructions, user profile information, file history, contact information, etc. Additionally, the site **302** can include advertisements, promotions, coupons, and other offers **304** available for the user **104** to utilize. As example, FIG. **3A** illustrates a user **104** browsing site **302** and in particular, user **104** reading an article on creating a blog. During the preview, an offer from a merchant (e.g., East) presenting a coupon **304** with a discount and/or free shipping when a purchase is made.

Note that unlike a conventional system (as illustrated in FIG. **1**), coupon **304** presents the user with the option to “Add to Wallet,” or “Save to Wallet.” In the current embodiment, the Add to Wallet option enable a user to save the promotion for later use. That is to say, unlike conventional systems where the promotion presented is for use at the moment and re-routes the user to a new site, coupon **304** instead enables continual browsing of the current site **302** while saving the coupon **304** for later use.

FIG. **3B**, presents the payment provider site **304** where a user **104** can retrieve a coupon **304** that was added to wallet at an earlier point in time. As an example, a user **104** is logged into his/her PayPal credit site **306**, where in addition to the user information including balance, available credit, etc., the site also illustrates the available coupons the user has added and are available for use. In this instance, the user **104**, who was earlier reading about blogging can now retrieve the coupon(s) **304**. Upon logging into and/or authenticating at the provider site, as illustrated in FIG. **3B**, the save coupon **304** is presented and available for the user. Note that in some instances, the saved coupons **304** may be available on the home site of the provider, in a coupon tab/founder, etc.

At FIG. **3C**, an illustration is given of a user **104** is ready and able to use the saved coupon **304**. In particular, FIG. **3C** is illustrated where a user **104** is at the merchant site (e.g., East) **310** for purchasing a product. As an example, the user **104** has an interest in purchasing a pair of pants and has been offered free shipping **312** as available on with the saved coupon **304**. Note that the merchant site may be visited through the merchant application, via retrieval of the site **310** using an internet service provider, or via the selection of the saved coupon **304** from the provider site **306**.

Now turning to FIG. **3D**, the provider site **306** is again illustrated where other offers **356** are available and presented to the user **104** for selection and later use. Note that although the offer was presented to the user **104** from a site **302** while the user **104** was browsing, other offers may be available on a digital wallet, from a service provider application, from a merchant site, as well as other sites including but not limited to news sites, learning sites, during video streaming, from a social networking site, etc. In addition, note that the offers **304** are retrievable from any user device including but not limited to a smart phone, tablet, home computer, wearable device, etc.

FIG. **4** illustrates a timing diagram **400** of the interactions between a client and the service provider for the presentation and storage of a tokenized offer. In FIG. **4**, the presentation of a tokenized offer that may be presented to a user **104** and added to a wallet is illustrated. To begin the presentation, a client **402** initiates a communication with a server for an offer related token request **416**. In one exemplary embodiment, a client (e.g., East) contacts the payment service provider (e.g., PayPal) for the opportunity to present the offer. The initial contact can include a request for a token **416**. In response to the request **416**, the server **404** can communicate with a secondary customer service provider **406** or customer service entity, data structure, database or the like, for the verification **418** that the client **402** is an existing customer **418**. Additionally, customer verification may also occur at verification **418**. During this verification, a merchants restrictions and clients information may be checked to ensure that merchant preferences are met and the customer (e.g., user **104**) should be provided with the offer. That is to say, a check on whether the customer is an appropriately targeted customer. For example, a merchant may have a restriction indicating the preference for customers under 25

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years old. Verification of customer age may occur prior to offering the token. In addition, a verification that the customer is a customer of the payment provider (e.g., PayPal customer) may occur as well as check for any outstanding risk warnings with the payment provider. Further, a check of customer cookies for relevant advertisements, search history and/or profile information, relationships, and communication on other related applications (e.g., social media) can be considered for verification of the customer.

Once the customer service entity **406** confirms the customer/merchant, the payment provider (e.g., server **404**) can grab the offer details **420**, that will be surfaced to the customer via the client site. For example, the coupon that will be presented on the news site visited by the customer. The offer details may be obtained from a separate entity, offers service, or campaign service entity **408**. Note that the offer service entity **408** may be a separate system, data structure, database, or the like. The offer details can include the offer identification number, the monetary amount to be discounted, the underwriter (e.g., who pays for the offer—payment provider, merchant, etc.), offer dates, expiration, etc. In addition, if the merchant has a campaign or other relationship with the payment provider server **404**. The relationship can include a retainer amount that may be provided from the merchant to the payment provider for the use when presenting the offers to the customers. For example, a merchant can provide a payment provider like PayPal with an upfront monetary amount of \$500 to surface the offers to customers. Each time a customer clicks on the save to wallet button, a designated amount is pulled from that account. If the offer is used then the funds are used, else if the offer is not used and/or expires then the funds can be returned for use with another customer. Alternatively, if an agreement/campaign does not exist, the payment provider, merchant, or other financial institution may front a percentage or the full amount of the offer.

The designated amount that is pulled from the campaign amount is confirmed available and updated when the offer **422** is presented to the customer **422**. In one embodiment, the offer is presented and pushed by the client **402**. However, in some instances the offer may be pushed by the merchant, social media site or by other site of relevance to the customer. As illustrated in FIG. 4, one the offer details are verified, the offer is displayed **422** on the client site and if desired by the customer a button clicked **424** (e.g., “Save to Wallet”).

A customer (e.g., user **104**) interested in the offer through the use of the clicked button **424** will then initiate a communication between the server **404** and the offer services entity **408** for the update of funds **426**. The update of funds can include a reduction in the monetary amount held in the campaign amount, an update in a service provider or merchant account (if the amount is fronted by the provider and/or merchant), etc.

Once the funds have been updated, the server **404** can respond to the client **402** with a tokenized offer. The offer along with offer details can be stored in a digital wallet in the user device **102**, an internal or external storage unit, and/or other component from which the offer details can be retrieved from. If the customer is a customer of the payment provider (e.g., PayPal user), then the offer can be saved to the PayPal wallet and/or local data storage for local search. Alternatively, the offer can be saved to a local data storage for use and retrieval without the need to be a client, customer, or affiliated with a specific financial institution, payment provider, and/or merchant. Included in the token

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offer can be the offer details, expiration date, merchant, authenticating information, and other relevant information.

Once the offer token is transmitted **428** to the client, priority information is kept in the server **404** and may be stored in a separate entity such as a date priority queue **410**. The data priority queue **410** may prioritize the token and keep track of the corresponding expiration dates. For example, if a token to East is approaching expiration in two days, the customer **414** may be sent emails **432** updating on the soon expiring token. In some instance, in addition to or instead of an email notification, a message, a customized push notification from the merchant **414** and/or the payment provider server **404** (i.e., client and/or PayPal is notified that an offer for a client is soon expiring), an update from the client site, or the like may be sent as a reminder. In one embodiment, the offer may be a short timed offer which can include continuous or interval notifications to the customer **414** via the client site **402** (not shown). For example, a customer **414** may be presented with an offer that has a 30 minute life, notifications may appear as reminders on the client site as the user continues to browse. If the token expires and it was part of a campaign or other agreement, the funds originally pulled for the token may be returned for use with another token offer.

At a later instance, when the user **104** is ready to use the offer, the user **104** (customer **414**) may visit the merchant site or open their digital wallet, PayPal app, or other where the offer may be retrieved. In one embodiment, the user may open up the merchant site and the offer may be automatically applied to the purchase at checkout. In another embodiment, when the customer arrives at the merchant’s website, a banner, overlay, or notification can pop up reminding the customer to save the offer or that a saved offer is pending and ready to be used. Upon use of the offer, if the merchant **412** is tied to a campaign or other agreement with the payment provider, the offer amount may be reduced from the account or retainer amount. (not shown)

Note that the offers may be surfaced within any web experience, thus anywhere where an advertisement may exist, an offer may similarly exist that may be saved to a wallet. In addition, the offers are also available and applicable to the account at checkout. A rebate may also be presented with money back with purchase. Note that an offer may be provided to any user **104** without the need for a specific membership with the financial institution. In some embodiments, a user may checkout with retrieval of the offer that may be stored locally on the user device **102**. Alternatively, the user **104** may use the offer after confirmation of membership with the financial institution or payment provider. If the user **104** is not a member, onboarding may be initiated at checkout.

FIG. 5 illustrates example process **500** for implementing a system that provides tokenized offers as computing system **200** of FIG. 2 and/or timing diagram **400** of FIG. 4. In particular, FIG. 5 illustrates a flow diagram illustrating operations for verifying and presenting a customer with a tokenized offer over a client site for saving to a digital wallet. According to some embodiments, process **500** may include one or more of operations **502-518**, which may be implemented, at least in part, in the form of executable code stored on a non-transitory, tangible, machine readable media that, when run on one or more hardware processors, may cause a system to perform one or more of the operations **502-518**.

Process **500** may begin with operation **502**, where a system such a payment provider system and server receive a request for an offer from a client system for an associated

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site. The offer request received at the server may be affiliated with a merchant, service provider, charity, non-profit, or the like. The offer may come in the form of a coupon presented on the client site for promotions, rebates, discounts, incentives, points, rewards, etc. for use the a user at the merchant.

Once the request is received, the server may communicate with another system, entity, database, data structure or service for the verification of information. The information verification can include verification of the merchant information and customer information. Among other checks, the customer may be checked to verify that merchant restrictions are met, risks are checked, user preferences, search history, and social relationships are checked. The checks can help indicate whether the customer is a good candidate to who the offer should be presented.

If the customer does not meet the predefined restrictions set by the merchant at operation 504, then the offer request may be denied. A denied offer indicates to the client that the offer request may not be fulfilled and the site should not present this offer to the customer. In one embodiment, if the customer does not meet the predefined restrictions, other types of advertisements and even promotions may continue to be displayed on the client site. For example, a customer may be presented with advertisements and even offers that may not include restriction checks or include other restrictions that the current customer meets. Alternatively, if all checks/restrictions are met, then an offer may be displayed on the client site. An exemplary restriction set by the merchant may be age, as the merchant may be targeting an older audience.

At operation 510, a tokenized offer is present on the client site for selection by the user of the user device. Prior to the presentation of the offer, the server may collect offer information (e.g., savings, type, expiration, etc.) as well as check for any associated relationships, campaigns and agreements established by the merchant and the server. If a retainer exists, a predefined amount may be deducted from the account to account for the offer discounts. In some instances, a campaign may not exist in which case the offer amount may be paid for by the server in its entirety or partially as an agreement with the merchant. Further, the tokenized offer may be stored locally in the user device and in a digital wallet affiliated with the server, payment provider, financial institution, merchant, etc. Note that in some instances, the predefined amount may not be deducted until operation 516.

Upon presentation of the tokenized offer to the customer, the customer then has the opportunity to save to wallet. Wishes to forgo the offer, then the offer is not saved. In instances, where the offer is affiliated with a campaign, the monetary amount pulled from the account may be reimbursed for use with another offer. If, however, the user desires to use the offer and selects the button presented to save to wallet, then process 500 continues to operation 516. At operation 516, a token is generated for the offer for later retrieval (locally or remotely through digital wallet) and is prioritized by the server at operation 518. Also, at operation 516 if a retainer exists, a predefined amount may be deducted from the account to account for the offer discounts. In one embodiment, the token generated may be digitally signed by an underwriter (e.g., PayPal) such that it may not be tampered with by a customer. That is to say, if the data is tampered with, then the offer will not work. The token works as a unique code/signature that ties the offer to the customer such that customer targeting if available. Thus, a customer unique identifier is added to the token so that only the designated customer may redeem it. For a customer that does not have an affiliation with the financial institution,

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payment provider, merchant or the like, other tracking or unique identifiers may be added to the token. For example, the token may include, but is not limited to, a session identification number (ID), tracking ID, social media profile, etc., which can be used to uniquely identify the customer so that the token is not shared.

Prioritizing the offer at operation 518 enables the option to receive notifications via email, messaging, application updates, site notices, and the like, that the offer is soon expiring. The process 500 may then continue again when a user is ready to use and checkout at which point the tokenized offer is presented and applied.

FIG. 6 illustrates an example computer system 600 in block diagram format suitable for implementing on one or more devices of the system in FIG. 2. In various implementations, a device that includes computer system 600 may comprise a personal computing device 102 (e.g., a smart or mobile device, a computing tablet, a personal computer, laptop, wearable device, PDA, etc.) that is capable of communicating with a network 626. A service provider and/or a content provider may utilize a network computing device (e.g., a network server) capable of communicating with the network. It should be appreciated that each of the devices utilized by users 104, service providers, and content providers may be implemented as computer system 600 in a manner as follows.

Additionally, as more and more devices become communication capable, such as new smart devices using wireless communication to report, track, message, relay information and so forth, these devices may be part of computer system 600. For example, windows, walls, and other objects may double as touch screen devices for users 104 to interact with. Such devices may be incorporated with the systems discussed herein.

Computer system 600 may include a bus 610 or other communication mechanisms for communicating information data, signals, and information between various components of computer system 600. Components include an input/output (I/O) component 604 that processes a user action, such as selecting keys from a keypad/keyboard, selecting one or more buttons, links, actuatable elements, etc., and sending a corresponding signal to bus 610. I/O component 604 may also include an output component, such as a display 602 and a cursor control 608 (such as a keyboard, keypad, mouse, touchscreen, etc.). In some examples, I/O component 604 may include an image sensor for capturing images and/or video, such as a complementary metal oxide semiconductor (CMOS) image sensor, and/or the like. An audio input/output component 606 may also be included to allow a user 108 to use voice for inputting information by converting audio signals. Audio I/O component 606 may allow the user 104 to hear audio. A transceiver or network interface 622 transmits and receives signals between computer system 600 and other devices, such as another user device, a merchant server, an email server, application service provider, web server, a payment provider server, and/or other servers via a network. In various embodiments, such as for many cellular telephone and other mobile device embodiments, this transmission may be wireless, although other transmission mediums and methods may also be suitable. A processor 618, which may be a micro-controller, digital signal processor (DSP), or other processing component, that processes these various signals, such as for display on computer system 600 or transmission to other devices over a network 626 via a communication link 624. Again, communication link 624 may be a wireless communication in some embodiments. Processor 618 may also

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control transmission of information, such as cookies, IP addresses, images, and/or the like to other devices.

Components of computer system 600 also include a system memory component 614 (e.g., RAM), a static storage component (e.g., ROM), and/or a disk drive 616. Computer system 600 performs specific operations by processor 618 and other components by executing one or more sequences of instructions contained in system memory component 612. Logic may be encoded in a computer readable medium, which may refer to any medium that participates in providing instructions to processor 618 for execution. Such a medium may take many forms, including but not limited to, non-volatile media, volatile media, and/or transmission media. In various implementations, non-volatile media includes optical or magnetic disks, volatile media includes dynamic memory such as system memory component 612, and transmission media includes coaxial cables, copper wire, and fiber optics, including wires that comprise bus 610. In one embodiment, the logic is encoded in a non-transitory machine-readable medium. In one example, transmission media may take the form of acoustic or light waves, such as those generated during radio wave, optical, and infrared data communications.

Some common forms of computer readable media include, for example, hard disk, magnetic tape, any other magnetic medium, CD-ROM, any other optical medium, RAM, PROM, EPROM, FLASH-EPROM, any other memory chip or cartridge, or any other medium from which a computer is adapted to read.

Components of computer system 600 may also include a short range communications interface 620. Short range communications interface 620, in various embodiments, may include transceiver circuitry, an antenna, and/or waveguide. Short range communications interface 620 may use one or more short-range wireless communication technologies, protocols, and/or standards (e.g., WiFi, Bluetooth®, Bluetooth Low Energy (BLE), infrared, NFC, etc.).

Short range communications interface 620, in various embodiments, may be configured to detect other devices (e.g., primary device, secondary device, etc.) with short range communications technology near computer system 600. Short range communications interface 620 may create a communication area for detecting other devices with short range communication capabilities. When other devices with short range communications capabilities are placed in the communication area of short range communications interface 620, short range communications interface 620 may detect the other devices and exchange data with the other devices. Short range communications interface 620 may receive identifier data packets from the other devices when in sufficiently close proximity. The identifier data packets may include one or more identifiers, which may be operating system registry entries, cookies associated with an application, identifiers associated with hardware of the other device, and/or various other appropriate identifiers.

In some embodiments, short range communications interface 620 may identify a local area network using a short range communications protocol, such as WiFi, and join the local area network. In some examples, computer system 600 may discover and/or communicate with other devices that are a part of the local area network using short range communications interface 620. In some embodiments, short range communications interface 620 may further exchange data and information with the other devices that are communicatively coupled with short range communications interface 620.

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In various embodiments of the present disclosure, execution of instruction sequences to practice the present disclosure may be performed by computer system 600. In various other embodiments of the present disclosure, a plurality of computer systems 600 coupled by communication link 624 to the network (e.g., such as a LAN, WLAN, PTSN, and/or various other wired or wireless networks, including telecommunications, mobile, and cellular phone networks) may perform instruction sequences to practice the present disclosure in coordination with one another. Modules described herein may be embodied in one or more computer readable media or be in communication with one or more processors to execute or process the techniques and algorithms described herein.

A computer system may transmit and receive messages, data, information and instructions, including one or more programs (i.e., application code) through a communication link 624 and a communication interface. Received program code may be executed by a processor as received and/or stored in a disk drive component or some other non-volatile storage component for execution.

Where applicable, various embodiments provided by the present disclosure may be implemented using hardware, software, or combinations of hardware and software. Also, where applicable, the various hardware components and/or software components set forth herein may be combined into composite components comprising software, hardware, and/or both without departing from the spirit of the present disclosure. Where applicable, the various hardware components and/or software components set forth herein may be separated into sub-components comprising software, hardware, or both without departing from the scope of the present disclosure. In addition, where applicable, it is contemplated that software components may be implemented as hardware components and vice-versa.

Software, in accordance with the present disclosure, such as program code and/or data, may be stored on one or more computer readable media. It is also contemplated that software identified herein may be implemented using one or more computers and/or computer systems, networked and/or otherwise. Where applicable, the ordering of various steps described herein may be changed, combined into composite steps, and/or separated into sub-steps to provide features described herein.

The foregoing disclosure is not intended to limit the present disclosure to the precise forms or particular fields of use disclosed. As such, it is contemplated that various alternate embodiments and/or modifications to the present disclosure, whether explicitly described or implied herein, are possible in light of the disclosure. For example, the above embodiments have focused on merchants and customers; however, a customer or consumer can pay, or otherwise interact with any type of recipient, including charities and individuals. Thus, “merchant” as used herein can also include charities, individuals, and any other entity or person receiving a payment from a customer. Having thus described embodiments of the present disclosure, persons of ordinary skill in the art will recognize that changes may be made in form and detail without departing from the scope of the present disclosure. Thus, the present disclosure is limited only by the claims.

What is claimed is:

1. A method, comprising:

in response to receiving a first offer from a first entity, accessing user information associated with a user; determining, based on the user information, that characteristics of the user meet one or more conditions

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associated with one or more of a cookie associated with a web browser, an identifier (ID) associated with a user device of the user, or a user ID used for one or more past transactions of the user;

in response to determining that characteristics of the user meet the one or more conditions, causing the first offer to be presented via an interactive element on a display of the user device, wherein the first offer is presented during a user interaction, with a particular page of a website or an application provided by a second entity, on the user device; and

in response to the user interaction with the interactive element, associating a token corresponding to the first offer with the user, wherein the token is associated with the user without redirecting the user away from the particular page of the website or the application, wherein the token is signed by a third-party payment provider such that a tampering of the token will invalidate the first offer, wherein the token is usable in a subsequent transaction between the user and the first entity by applying the first offer to the subsequent transaction, wherein the subsequent transaction is conducted at least in part via a checkout interface provided by the first entity, wherein the first entity is different from the second entity that provides the particular page of the website or the application, and wherein using the token in the subsequent transaction comprises:

in response to determining that the user device is involved in the subsequent transaction with the first entity associated with the first offer, causing the first offer to be retrieved using the token; and

facilitating, based on a verification of a unique token signature of the token, the transaction at least in part by applying the first offer to the subsequent transaction.

2. The method of claim 1, further comprising:

identifying the first offer in response to said determining that the characteristics of the user meet the one or more conditions; and

in response to said identifying the first offer, associating the first offer with a user account of the user.

3. The method of claim 1, wherein in response to said determining that the characteristics of the user meet the one or more conditions, the method further comprises generating, based on one or more of the first offer and the user, the token for the first offer and associating the token with a user account of the user.

4. The method of claim 1, wherein the unique token signature is also unique to one or more of: the first offer, a session identifier of a session associated with the user interaction with the website or the application, or a tracking ID associated with the user or the user device.

5. The method of claim 1, wherein the first offer includes one of: a promotion, points, a discount, or a coupon that can be applied in the subsequent transaction with the first entity via a user account of the user.

6. The method of claim 1,

wherein said associating the token comprises causing the token to be saved in a local data storage of the user device accessible by a digital wallet of the user; and

wherein the token is usable in the subsequent transaction by retrieving the token from the local data storage of the user device and processing the subsequent transaction using the token.

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7. The method of claim 1, wherein the token is usable in the subsequent transaction between the user and the second entity without establishing a user account, for the user, with the second entity.

8. The method of claim 1, wherein the first offer is selected from a plurality of offers from one or more first entities including the first entity based on said determining that the characteristics of the user meet one or more conditions that are associated with the first offer.

9. The method of claim 1, wherein after the token corresponding to the first offer is associated with the user, a visual representation of the first offer is viewable in a website or an application of the third-party payment provider.

10. A server system, comprising:

a non-transitory memory storing instructions; and

a processor configured to execute the instructions to cause the server system to:

access a plurality of offers from a plurality of entities, each of the offers associated with one or more respective restrictions for potential customers;

determine, based on user information for a certain customer, that the certain customer meets one or more certain restrictions for a certain offer, of a first entity, of the plurality of offers;

based on a determination that the certain customer meets the one or more certain restrictions for the certain offer, cause the certain offer to be presented via an interactive element on a display of a user device of the certain customer, wherein the certain offer is presented during a user interaction, with a particular page of a website or an application provided by a second entity, on the user device; and

based on the user interaction with the interactive element, associate a token corresponding to the certain offer with the certain customer while the particular page continues to be presented, wherein the token is configured to prevent tampering thereof and is only redeemable by the certain customer, and wherein the token is usable in a subsequent transaction between the certain customer and the first entity at least in part by:

causing the certain offer to be retrieved using the token based on a determination that the user device is involved in the subsequent transaction with the first entity, the subsequent transaction conducted at least in part via a checkout page provided by the first entity, wherein the first entity is different from the second entity that provides the particular page of the website or the application; and

processing, based on a verification of a unique token signature of the token, the subsequent transaction at least in part by applying the certain offer to the subsequent transaction.

11. The server system of claim 10, wherein executing the instructions further causes the server system to,

based on the determination that the certain customer meets the one or more certain restrictions for the certain offer, generate, based on one or more of the certain offer and the certain customer, the token for the certain offer and associate the token with a user account of the certain customer.

12. The server system of claim 10, wherein the unique token signature is also unique to one or more of: the certain offer, a session identifier (ID) of a session associated with

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the user interaction with the website or the application, or a tracking ID associated with the certain customer or the user device.

13. The server system of claim 10, wherein the certain offer includes one of: a promotion, points, a discount, or a coupon that can be applied in the subsequent transaction with the first entity.

14. The server system of claim 10, wherein the token is associated with the certain customer by being saved in a local memory of the user device of the certain customer; and wherein the token is usable in the subsequent transaction by retrieving the token from the local memory of the user device and processing the subsequent transaction using the token.

15. The server system of claim 10, wherein after the token corresponding to the certain offer is associated with the certain customer, a visual representation of the certain offer is viewable in a website or an application associated with the server system.

16. A non-transitory machine readable medium having stored thereon machine readable instructions executable to cause a machine to perform operations comprising:

accessing a plurality of offers from a plurality of entities, each of the offers associated with one or more respective restrictions for potential users;

matching, based on user information for a certain user, the certain user with a certain offer of the plurality of offers, wherein the certain offer is from a first entity, and wherein one or more characteristics of the certain user match one or more certain restrictions for the certain offer;

causing the certain offer to be presented via an interactive element on a display of a user device of the certain user, wherein the certain offer is presented during a user interaction, with a particular portion of a website or an application associated with a second entity, on the user device;

associating, based on the user interaction with the interactive element and without navigating to a different portion of the website or the application, a token corresponding to the certain offer with the certain user,

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wherein the token is signed in a manner to prevent a tampering of the token, and wherein the token is usable in one or more subsequent transactions between the certain user and the first entity;

based on a determination that the certain user has initiated a transaction of the one or more subsequent transactions with the first entity via a checkout interface provided by the first entity that is different from the second entity associated with the website or the application, retrieving the token; and

processing, based on a verification of a unique token signature of the token, the transaction at least in part by applying the certain offer associated with the token to the transaction.

17. The non-transitory machine readable medium of claim 16, wherein the unique token signature is also unique to one or more of: the certain offer, a session identifier of a session associated with the user interaction with the website or the application, or a tracking ID associated with the certain user or the user device.

18. The non-transitory machine readable medium of claim 16, wherein the certain offer includes one of: a promotion, points, a discount, or a coupon that can be applied in the transaction with the first entity via an account of the certain user.

19. The non-transitory machine readable medium of claim 16,

wherein said associating the token comprises causing the token to be saved in a local storage of the user device that is accessible by a digital wallet of the certain user; and

wherein the token is usable in the one or more subsequent transactions by retrieving the token from the local storage of the user device and processing the one or more subsequent transactions using the token.

20. The non-transitory machine readable medium of claim 16, wherein after the token corresponding to the certain offer is associated with the certain user, a visual representation of the certain offer is viewable in a digital wallet of the certain user.

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